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COMMENT

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Defining conceptual boundaries of knowledge attitudes and practices to improve validity in public health research

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Abstract

Research based on the Knowledge–Attitudes–Practices (KAP) framework has substantially contributed to the understanding of health-related behaviors at the population level and has been widely used in the design and evaluation of public health programs and policies. However, conceptual precision in the definition and measurement of its domains is critical to preserving the validity and interpretability of findings. This comment examines a recurrent methodological limitation in KAP-based research: the lack of clear conceptual boundaries between knowledge, attitudes, and practices. Drawing on recent examples from different areas of health research, it illustrates how the inclusion of heterogeneous content within a single domain—including knowledge, behaviors, intentions, and contextual factors—can lead to conceptual overlap, misclassification bias, and difficulties in interpreting relationships between variables. It further highlights that psychometric procedures, such as factor analysis, do not necessarily resolve these ambiguities when a clear conceptual foundation is lacking in instrument development. We argue that strengthening the conceptual and methodological coherence of the KAP framework—through explicit definitions, clear criteria for item classification, and appropriate validation processes—is essential to improve comparability across studies and the overall quality of the evidence generated. In this context, a precise delineation of KAP domains is crucial for producing more robust and actionable evidence to inform public health interventions, strategies, and policy development.

Keywords KAP models, Measurement validity, Misclassification bias, Behavioral measurement, Public health

1 Main text

Research based on the knowledge–attitudes–practices (KAP) framework has a long tradition in public health and in the design and evaluation of programs and policies. This approach has been widely used to study health-related behaviors across diverse contexts, including nutrition [1, 2], vitamin D supplementation and sun exposure [3, 4], vaccine acceptance and uptake [5, 6], physical activity [7], addiction-related behaviors [8], and other public health domains [9–11].

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Beyond the academic sphere, the KAP framework has been adopted by international organizations and institutional actors for the design of interventions and the generation of applied evidence. In this context, methodological guidelines have been developed by the World Health Organization (WHO) [12] and the Food and Agriculture Organization (FAO) of the United Nations [13], and the framework has also been used in programs implemented by governments and nongovernmental organizations, particularly in areas related to childhood and community health [14, 15].

These applications have significantly contributed to improving the understanding of how knowledge and attitudinal factors are related to preventive practices, health behaviors, and population risk profiles. This evidence is particularly valuable for informing public health strategies in contexts where behavioral determinants play a central role in disease prevention and health promotion.

Early studies based on the KAP framework, such as that of Schwartz [1], sought to establish a clear taxonomic distinction among the three domains of the model. Knowledge was operationalized through factual validity, assessing the understanding of scientific concepts and nutritional principles. Attitudes were defined and measured as emotional predispositions to respond in a particular way to a given object using subjective rating scales. Finally, practices were operationalized through the frequency of observable actions and concrete habits, distinguishing between the personal domain (individual dietary habits) and the professional domain (the frequency of incorporating educational activities into their work). For Schwartz, attitudes were understood as learned, emotionally toned predispositions to respond in a particular way toward a given object, whereas practices were operationalized as the reported frequency of specific behaviors. Within this framework, maintaining a clear distinction between these domains—between evaluative predispositions and observable actions—is a sine qua non condition for the structure of the model.

However, in more recent guidelines and empirical applications, the constructs of the KAP framework are rarely defined conceptually. Instead, they are often described operationally as what "people know, how they feel, and how they behave in relation to a particular topic" [14, p.5] or as a way to collect information on "what is known, believed and done" [12, p.6], or more broadly as "awareness, beliefs, and behaviors related to health" [2, p.2], with limited theoretical grounding and unclear boundaries between domains. In this sense, the KAP framework appears to have reached a high degree of methodological standardization while continuing to exhibit insufficient conceptual development. Methodological standardization refers to the recurrent use of similar survey structures, scoring systems, and validation procedures across KAP studies, whereas insufficient conceptual development refers to the lack of theoretical clarification regarding the boundaries between knowledge, attitudes, and practices. These limitations have been acknowledged even in key reference documents, such as the FAO manual, which states that KAP studies "usually cannot be compared because of major differences in study design…and in how knowledge, attitudes and practices are defined and measured" [13, p.1].

Despite its widespread use, the KAP framework rests on the need for a clear conceptual delineation among the three fundamental domains of the model: knowledge, attitudes, and practices. In this regard, it is necessary to further clarify the distinction between these components, considering that knowledge refers to factual or scientific

cognitive content, attitudes refer to subjective evaluative predispositions, and practices refer to concrete and observable behaviors. The lack of a precise differentiation between these domains may compromise the validity of inferences derived from KAP analyses, particularly when measurement items do not adequately reflect the nature of the domain they are intended to assess.

In this context, relevant methodological considerations emerge that may strengthen future applications of the KAP framework in public health research. In particular, one of the most frequent issues concerns the inappropriate classification of behavioral items within the attitudinal domain. For example, statements such as "I often use a parasol (sunshade or umbrella) and sunscreen lotion to prevent sunshine" [3] assess habitual actions rather than subjective dispositions, as their wording reflects behavioral frequency ("I often use"), a characteristic of the practices domain. In contrast, statements such as "I like to expose to sunlight" [3, 4] more appropriately represent an individual's preference or evaluation, which is consistent with the classical definition of attitude. This distinction illustrates the fundamental conceptual difference between attitudes (evaluative predispositions) and practices (observable behaviors) within the KAP model.

The inclusion of behaviors within the attitudinal domain introduces a conceptual overlap between attitudes and practices, which is further reinforced when similar items appear simultaneously in both domains. For example, the absence of sunscreen use is also classified as a practice ("Not using Sun Protective Factor (SPF) contain creams") [3, 4], generating redundancy in measurement. This type of overlap may induce misclassification bias, artificially inflate associations between domains because of shared content, and hinder the interpretation of results. In this sense, a clear differentiation between attitudes (preferences and evaluative dispositions) and practices (behaviors actually performed) is essential to preserve the internal validity of KAP-based studies.

In line with the above, some recent studies classify attitudes as a heterogeneous set of statements that depart from the classical conceptualization of the construct. In the D-KAP-38 questionnaire [16], for example, attitudinal items include structural conditions of the environment—such as urbanization, lack of public spaces, or indoor work schedules—as well as educational, cultural, and economic barriers; factual beliefs (e.g., the presumed greater effectiveness of supplements compared to diet or sun exposure); normative judgments; and descriptions of third-party behaviors. Taken together, these items do not necessarily capture individuals' evaluative predispositions but rather a combination of contextual perceptions, causal explanations, and informational content, thereby blurring the conceptual boundaries of the attitudinal domain.

This type of ambiguity is also observed within a single instrument. For example, the variable "Exposure to sunlight" is classified as an attitude, despite its behavioral nature or, at least, its ambiguity between disposition and behavior. Moreover, the practices domain includes items such as "Physician prescribed Vitamin D with specific dosage" and "Willing to take Vitamin D regularly" [17]. The former does not represent a practice performed by the individual but rather an action carried out by a healthcare professional, placing the respondent in a passive role. The latter reflects an intention or future disposition rather than an observable behavior. These examples illustrate how, within the same instrument, attitudes, intentions, individual behaviors, and third-party actions may coexist, thereby blurring the boundaries between domains and complicating the interpretation of results.

Finally, these inconsistencies are not limited to a specific thematic area but are observed across different fields of health. In the context of vaccination, for example, a validation study among healthcare students proposes a conceptual distinction between knowledge, attitudes, and practices but reports that several attitudinal items show low or even negative internal consistency and are therefore removed from the instrument, suggesting difficulties in the operationalization of the attitude construct. However, despite the use of statistical validation techniques such as factor analysis, items such as "What other vaccines have you received" and "Are you aware that several vaccines require booster doses for full protection" [5] are ultimately retained within the attitudinal domain, with the former referring to past behavior and the latter to knowledge about vaccination. This example suggests that statistical procedures do not necessarily resolve prior conceptual ambiguities and that overlap between domains may persist even after formal validation processes.

Similarly, in a study on human papillomavirus vaccination, the authors indicate that the attitudinal domain is composed of items 24 to 32 of the instrument. Within this set, statements such as "Vaccines prevent disease", "Generally, I recommend vaccines to patients who meet the eligibility criteria", "HPV vaccine prevents cervical cancer" and "I recommend the HPV vaccine to patients who are eligible" [6] are included. Taken together, these findings indicate that conceptual ambiguity in KAP applications is reflected not only in the classification of individual items but also in the inconsistent redefinition of domains, thereby compromising comparability across studies and the validity of the inferences derived.

The inconsistencies observed in the delineation of KAP domains have direct implications for the validity of studies. First, conceptual overlap between knowledge, attitudes, and practices may introduce misclassification bias by assigning items to domains that do not correspond to their theoretical nature. Second, the inclusion of shared content across domains (for example, behaviors framed as attitudes) may artificially inflate associations between them, thereby compromising the interpretation of relationships such as knowledge–attitude–practice. This may lead to erroneous conclusions about causal relationships between domains by attributing effects to constructs that have not been measured independently. More broadly, this overlap may introduce dependency between domains, making it difficult to distinguish their specific effects. Third, the incorporation of intentions, subjective norms, or perceived control within the practices domain blurs the distinction between observed behavior and its determinants, thereby affecting construct validity.

Taken together, these considerations suggest that, although the KAP framework remains widely used, its effective application depends on greater conceptual and methodological clarity—a necessary condition for improving the validity, comparability, and practical utility of evidence in public health.

Based on the above, a set of recommendations can be outlined to strengthen the use of the KAP framework in applied research. First, an explicit conceptual delineation of the three domains is needed, distinguishing knowledge (factual information), attitudes (individual evaluative predispositions), and practices (observable behaviors actually performed). This distinction constitutes a necessary condition to avoid conceptual overlap and to preserve measurement validity.

Second, in line with these criteria, it is recommended that items be evaluated by expert judges prior to their application, with the aim of verifying their alignment with the corresponding theoretical domain. The evaluation and classification of items may be supported by inter-rater agreement methods, such as the Kappa coefficient, allowing for the quantification of consistency in item classification and the identification of conceptual ambiguities at early stages of instrument development. In practical terms, it is recommended that attitudinal items avoid action verbs or frequency adverbs (e.g., always, frequently) and be formulated as subjective evaluations (e.g., I consider that…, I find it important that…), ensuring that the measurement of behavior remains exclusively within the practices domain.

Third, although factor analyses can provide evidence on the empirical structure of instruments, their use should be interpreted with caution in the absence of a clear conceptual foundation. In this regard, it is advisable to first incorporate analyses of internal consistency and item performance, as in recent validation studies, where items with low homogeneity or inconsistent correlations are identified and, when appropriate, excluded prior to proceeding with more complex analyses [5]. This type of preliminary evaluation may contribute to improving the overall quality of the instrument, although it does not, by itself, guarantee the conceptual validity of the domains, as ambiguities in item classification may persist. In this sense, such preliminary assessment allows for the identification of definitional or classification issues that cannot be resolved solely through statistical techniques. In addition, future KAP research focused on behavioral outcomes may benefit from incorporating objective measures, such as digital tracking devices or AI-assisted behavioral detection tools, to complement self-reported practices.

Finally, moving toward more consistent methodological standards, including explicit operational definitions, transparent criteria for item classification, and detailed reporting of validation processes, is recommended to improve comparability across studies and the validity of inferences derived from the KAP framework.

2 Conclusion

A clear conceptual delineation between knowledge, attitudes, and practices is essential for the validity and interpretability of studies based on the KAP framework. The lack of precise differentiation between these domains may introduce misclassification bias, hinder the interpretation of relationships between variables, and compromise comparability across studies. In this context, strengthening the conceptual and methodological coherence in the definition and measurement of KAP domains emerges as a necessary condition for improving the quality of the evidence generated and its usefulness in the design of public health strategies, health education programs, and policies aimed at disease prevention and health promotion.

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